# Textbook form of the Frohman problem

Benjamin Stanley Frohman

Copyright (c) 2026 Benjamin Stanley Frohman. Apache-2.0.

https://github.com/BenFrohman/Frohman-MixedCorrelators-FJ

**Problem (Frohman, mixed correlators of a three-block chain sum).**

Let $W = u^5 v + v^6$ and $F = W \oplus W \oplus W$. Let $J$ scale every coordinate of $\mathbb{C}^6$ by a primitive sixth root of unity. The pair $(F, \langle J \rangle)$ is an admissible Landau–Ginzburg orbifold with $\hat{c}(F) = 4$.

The FJRW state space of $(F, \langle J \rangle)$ has dimension $2610$: one broad sector of dimension $2605$ and five narrow Lefschetz states of dimension $1$.

Compute every genus-zero primary three-point number

$$\langle \alpha, \beta, \gamma \rangle_{0,3}^{F,\langle J \rangle} = \int_{\overline{\mathcal{M}}_{0,3}^{F,\langle J \rangle}} \left[\overline{\mathcal{M}}_{0,3}^{F,\langle J \rangle}\right]^{\mathrm{vir}}$$

where $\alpha, \beta, \gamma$ run over that $2610$-dimensional space, including mixed broad–narrow triples and mixed-block tensors, subject to the selection rule that decorations multiply to $J$.

Show that these numbers are not a product of three Guéré integrals of $W$. Produce the table, or prove vanishing, for every allowed triple.

That is the Frohman problem. It is not Guéré's theorem.